25901105 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBTS200 - ENZYMOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Enzymes that catalyze intramolecular rearrangements are known as: [U] / [CO1]

   a). Lyases b). Isomerases

   c). Ligases d). Transferases

2. Why are protease inhibitors often added during enzyme extraction? [U] / [CO3]

   a). To increase enzyme activity b). To prevent enzyme degradation by proteases

   c). To denature unwanted proteins d). To change enzyme kinetics

3. The main goal of enzyme purification is to: [U] / [CO3]

   a). Increase the enzyme concentration only b). Isolate the enzyme from other cellular components while retaining activity

   c). Denature unwanted proteins d). Change the enzyme’s specificity

4. LDH is a marker for [K] / [CO4]

a). Prostrate cancer b). Liver dysfunction

c). Heart Disease d). none of the above

5. Which of the following statements about proenzymes (zymogens) is TRUE? [U] / [CO1]

a). They are active immediately after synthesis b). They require proteolytic cleavage for activation

c). Their activation is always reversible d). They function as cofactors

6. Reciprocal regulation by hormones in glycogen metabolism is attributed to [K] / [CO2]

a). Allosteric regulation b). Covalent regulation c). Enzyme inhibition d). None of the above

7. A competitive inhibitor is added to an enzyme. It affects activation energy with respect to enzyme activity by [A] / [CO2]

a). Decreasing the activation energy and enhancing substrate binding

b). Increasing the activation energy for substrate conversion

c). Having no effect on activation energy

d). Permanently lowering the activation energy

8. A coupled enzyme assay is used when: [U] / [CO3]

a). Substrate is colored b). Enzyme is inactive

c). Product is difficult to detect directly d). Reaction is very slow

9. Mixed preparations of lipase,amylase and protease are used to treat [U] / [CO4]

a). Heart b). Lungs

c). Pancreas d). none of the above

10. What is the function of SDS in SDS-PAGE? [U] / [CO3]

a). To increase protein solubility

b). To provide uniform negative charge and denature proteins

c). To stain proteins

d). To maintain pH

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Define enzymes and list any two general properties of enzymes. [U] / [CO1]

12. What are serine proteases ? Mention their role in protein digestion. [K] / [CO2]

13. List out the various biophysical techniques used in enzymology. [U] / [CO3]

14. What are the roles of amylase and lipase in clinical biochemistry ? [K] / [CO4]

15. Give a short account on differentiating isoenzymes. [U] / [CO4]

16. Comment on X-ray Crystallography. [U] / [CO3]

17. What is FAD? Describe its role in enzyme catalysis. [K] / [CO2]

18. What is ultracentrifugation? Explain. [U] / [CO3]

19. What are the applications of lipase enzyme? [K] / [CO4]

20. Write short notes on competetitive inhibition. [K] / [CO2]

21. How do enzymes speed up chemical reactions? [U] / [CO1]

22. List the goals of studying enzyme kinetics. [An] / [CO2]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Explain Emil Fischer’s lock-and-key hypothesis and its historical importance. [U] / [CO1]

24. Explain cooperativity and feedback inhibition in allosteric enzymes. [K] / [CO2]

25. Write a detailed note on thin-layer chromatography (TLC) and its advantages over paper chromatography. [U] / [CO3]

26. Write a short note on active site and its mapping. [U] / [CO1]

27. Explain the salient features of active site and any two mapping methods of active site. [K] / [CO2]

28. Describe the various methods of enzyme extraction from cells. Discuss mechanical, chemical, and enzymatic approaches, with examples. [U] / [CO3]

29. Highight the major fermentation based products in food industry. [K] / [CO4]

30. Give a detailed account on holoenzyme. [U] / [CO1]

[5x6 = 30]